

LECTURE SHEET

EXERCISE-I

(Types of solids and Crystal Structures)

LEVEL-I (MAIN)

Straight Objective Type Questions

- Covalent solid among the following is
1) solid Ar 2) MgO 3) Fe 4) BN
- (A) : Glass possess sharp melting point.
(R) : Glass is a pseudo solid.
1) Both (A) and (R) are true and (R) is the correct explanation of (A)
2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
3) (A) is true but (R) is false 4) (A) is false but (R) is true
- Graphically the total number of fundamental spatial arrangements possible are
1) 3 2) 7 3) 10 4) 14
- In case of a cubic system, the number of types of space lattices are
1) 3 2) 7 3) 14 4) 12
- How many kinds of primitive unit cells are possible
1) 23 2) 7 3) 230 4) 14
- (Unitcell) (no of atoms per unitcell)
A) Simple cube 1) 4
B) fcc 2) 2
C) bcc 3) 1
The correct match is

A	B	C	A	B	C	A	B	C	A	B	C
1) 2	3	1	2) 2	1	3	3) 3	1	2	4) 1	2	3
- The crystal system without any element of symmetry is
1) monoclinic 2) hexagonal 3) triclinic 4) cubic
- White Sn belongs to one of the seven crystal systems. The number of Bravais lattices possible for that crystal system
1) 2 2) 1 3) 4 4) 3
- In KMnO_4 the crystallographic parameters are
1) $\alpha = \beta = \gamma \neq 90^\circ$ 2) $\alpha = \beta = \gamma = 90^\circ$ 3) $\alpha \neq \beta \neq \gamma \neq 90^\circ$ 4) $\alpha = \gamma = 90^\circ$; $\beta > 90^\circ$
- Copper metal belongs to a crystal system represented by the crystal dimensions as
1) $\alpha = \beta = \gamma = 90^\circ$; $a = b = c$ 2) $\alpha \neq \beta \neq \gamma$; $a = b = c$
3) $\alpha = \beta = 90^\circ$, $\gamma \neq 90^\circ$; $a = b = c$ 4) $\alpha = \beta = \gamma = 90^\circ$; $a \neq b \neq c$
- For a covalent solid, the units which occupy lattice points are
1) Atoms 2) Ions 3) Molecules or atoms 4) Electrons
- Which of the following does not give any diffraction bands with X-rays ?
1) BaSO_4 2) Graphite 3) Diamond 4) Plastic

13. The ratio of 'd' values in NaCl crystal is
 1) 0.703 : 1 : 1.134 2) 1 : 0.703 : 1.134 3) 1 : 1.134 : 0.703 4) 0.703 : 1.134 : 1
14. In x-ray diffraction experiment at which one of the following path difference between the two waves, destructive interference is observed (λ = wavelength of x-rays)
 1) λ 2) 2λ 3) 3λ 4) 1.5λ
15. A match box exhibits
 1) Cubic geometry 2) monoclinic geometry
 3) orthorhombic geometry 4) tetragonal geometry

Numerical Value Type Questions

16. For a crystal the angle of diffraction (2θ) is 90° and the second order line has a 'd' value of $2.28\text{\AA}$. The wave length (in $\text{\AA}$) of x-rays used for Bragg's diffraction is :
17. The number of space lattices possible for the crystallographic dimensions $\alpha \neq \beta \neq \gamma$.
18. At what angle for a first order diffraction, the distance between two adjacent planes of crystal is equal to the wavelength of x-rays used

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. At very low temperature, oxygen O_2 , freezes and forms a crystal. Which term best describes the solid
 a) Covalent network b) Molecular crystals c) Metallic d) Ionic
2. As it cools, olive oil slowly solidifies and forms a solid over a wide range of temperature. Which term best describes the solid?
 a) Ionic b) Covalent network c) Metallic d) Amorphous
3. Metals are good conductors of electricity because they contain
 a) ionic bonds b) a network structure
 c) very few valence electrons d) free electrons
4. In a metallic crystal
 a) the valence electrons constitute a sea of mobile electrons
 b) the valence electrons are localised in between the kernels
 c) the valence electrons remain within the field of influence of their own kernels
 d) none of the above
5. Triclinic crystal has the following unit cell parameters:
 a) $a = b = c$; $\alpha = \beta = \gamma = 90^\circ$ b) $a = b \neq c$; $\alpha = \beta = \gamma = 90^\circ$
 c) $a \neq b \neq c$; $\alpha \neq \beta \neq \gamma \neq 90^\circ$ d) $a = b \neq c$; $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$
6. The unit cell with parameters $\alpha = \beta = \gamma = 90^\circ$ and $a = b \neq c$ is
 a) Cubic b) Triclinic c) Hexagonal d) Tetragonal
7. In a hexagonal crystal:
 a) $\alpha = \beta = \gamma \neq 90^\circ$; $a = b = c$ b) $\alpha = \beta = \gamma = 90^\circ$; $a = b \neq c$
 c) $\alpha = \beta = \gamma = 90^\circ$; $a \neq b \neq c$ d) $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$; $a = b \neq c$
8. Orthorhombic crystal has the following unit cell parameters:
 a) $a = b = c$; $\alpha = \beta = \gamma = 90^\circ$ b) $a = b \neq c$; $\alpha = \beta = \gamma = 90^\circ$
 c) $a \neq b \neq c$; $\alpha = \beta = \gamma = 90^\circ$ d) $a = b \neq c$; $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$

9. The crystal system of a compound with unit cell parameters, $a = 0.328 \text{ nm}$, $b = 0.328 \text{ nm}$, $c = 0.527 \text{ nm}$ and $\alpha = \beta = \gamma = 90^\circ$ is
 a) Cubic b) Tetragonal c) Monoclinic d) Rhombohedral

More than One correct answer Type Questions

10. Glasses and plastics are
 a) amorphous solids b) supercooled liquids c) anisotropic d) ferromagnetic
11. If the three interaxial angles defining the unit cell are all equal in magnitude, the crystal cannot belong to the
 a) monoclinic system b) cubic system c) hexagonal system d) triclinic system
12. Which of the following solids fuse at moderate temperature?
 a) Ga_2 b) SiO_2 c) Solid CO_2 d) SiC
13. Which systems have more than one unit cell?
 a) Triclinic b) Hexagonal c) Monoclinic d) Orthorhombic

Linked Comprehension Type Questions

Passage :

When a metallic element combines with a non-metallic element, generally an electrovalent substance is formed. Metal atom loses electron or electrons to form cations. Non metal atom gains electrons to form anion. The oppositely charged ions are held together in an ionic solid by electrostatic attraction forces.

An example of formation of electrovalent substance: $\text{K(s)} + 1/2 \text{Br}_2(\text{l}) \rightarrow \text{KBr(s)}$

14. During melting of potassium bromide
 a) Ions are formed b) Ions are separated
 c) ions are made volatile d) Molecules are separated
15. Potassium bromide in its native state does not conduct electricity because
 a) It has no electrons b) It has no ions
 c) Positive charge of cation is nullified by negative charge of anions
 d) Ions do not move in the systematic lattice of the solid

Matrix Matching Type Questions

- | | |
|--|----------------------------------|
| 16. Column-I (Crystalline solids) | Column-II (Type) |
| A) Solid CO_2 | p) Hydrogen bonded structure |
| B) Silica | q) Conducting covalent solid |
| C) Graphite | r) Non-conducting covalent solid |
| D) Ice | s) Molecular crystalline solid |
| 17. Column-I (Nature of bonding) | Column-II (Material) |
| A) Metallic | p) Carborundum, silicon |
| B) Covalent | q) MgO |
| C) Vanderwaal's | r) Solid N_2 |
| D) Ionic | s) Sodium |

Integer Type Questions

18. How many unit crystal systems contain only one unit cell?
19. What is the maximum number of unit cells may be present in any crystal system?

EXERCISE-II

(Close Packed Structures and Voids)

LEVEL-I (MAIN)

Straight Objective Type Questions

1. Coordination number for Cu is
1) 1 2) 6 3) 8 4) 12
2. Which of the following is an example of body centred cube?
1) Mg 2) Zinc 3) Copper 4) Potassium
3. The co-ordination number of a metal crystallising in a hexagonal close packed structure is
1) 12 2) 4 3) 8 4) 6
4. In a cubic close packed structure the number of nearest neighbours for a given lattice point is
1) 6 2) 8 3) 12 4) 14
5. Which of the following structure is most uncommon for metals?
1) simple cubic 2) B.C.C. 3) C.C.P. 4) H.C.P.
6. A point that is located at the corner of a unit cell is shared by how many unit cells?
1) 2 2) 4 3) 6 4) 8
7. The number of lattice points per unit cell in B.C.C and end centered lattice respectively
1) 6, 6 2) 9, 10 3) 6, 8 4) 6, 10
8. A body centered cubic solid is made up of two elements A and B. Atoms of 'A' occupy two corners of the cube. Remaining positions of the unit cell are occupied by the atoms of 'B'. The formula of the compound is :
1) A_4B_7 2) A_7B_4 3) AB_7 4) A_7B
9. A compound formed by elements X and Y crystallizes in a cubic structure in which the X atoms are at the corners of a cube and the Y atoms are at the face-centres. The formula of the compound is
1) X_3Y 2) XY 3) XY_2 4) XY_3
10. The void between two oppositely directed planar triangles of spheres in adjacent layers is called
1) Cubic void 2) Tetrahedral void
3) Octahedral void 4) Tetrahedral (or) Octahedral void
11. How many Cl^- ions are there around Na^+ ion in NaCl crystal
1) 3 2) 4 3) 6 4) 8
12. Which of the following has hcp crystal structure?
1) NaCl 2) $CaCl_2$ 3) Zn 4) RbCl
13. How many atoms are there in a cube based unit cell having one atom on each corner and two atoms on each body diagonal?
1) 8 2) 6 3) 9 4) 4
14. A solid has a structure in which W atoms are located at the corners of the cubic lattice, O atoms at the centre of the edges and Na atom at the centre of the cube. The formula of the compound is
1) $NaWO_2$ 2) Na_2WO_3 3) $NaWO_3$ 4) $NaWO_4$

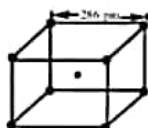
15. The number of octahedral voids in a unit cell of cubic close packed structure is
 1) 1 2) 2 3) 4 4) 8
16. (A) : A void surrounded by a triangle of spheres capped by another sphere is called tetrahedral void.
 (R) : Tetrahedral voids are in tetrahedral arrangement.
 1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
 3) (A) is true but (R) is false 4) (A) is false but (R) is true
17. In a face centred cubic arrangement of A and B atoms whose A atoms are at the corner of the unit cell and B atoms at the face centred. One of the A atom is missing from one corner in unit cell. The simplest formula of compound is :
 1) A_7B_3 2) AB_3 3) A_7B_{24} 4) A_3B_3
18. Lithium selenide can be described as a cubic closest-packed array of selenide ions with lithium ions in all of the tetrahedral holes. Formula of lithium selenide is
 1) Li_2Se 2) $LiSe$ 3) $LiSe_2$ 4) Li_3Se
19. In the closest packing of atom A, the radius of atom B that can be fitted into tetrahedral void is
 1) $r_B = 0.155r_A$ 2) $r_B = 0.225r_A$ 3) $r_B = 0.414r_A$ 4) $r_B = 0.732r_A$
20. What is the co-ordination number of sodium in Na_2O ?
 1) 6 2) 4 3) 8 4) 2
21. Total volume of atoms present in a face centered cubic unit cell of a metal is
 1) $\frac{24}{3}\pi r^3$ 2) $\frac{12}{3}\pi r^3$ 3) $\frac{16}{3}\pi r^3$ 4) $\frac{20}{3}\pi r^3$

Numerical Value Type Questions

22. The minimum radius ratio that can give a specific coordination number to the compound is
23. Sodium metal crystallises in a body-centred cubic lattice with the cell edge, 'a' = 4.29 Å. The radius of the Na-atom will be (Å)
24. For an octahedral arrangement the lowest radius ratio limit is

LEVEL-II (ADVANCED)**Straight Objective Type Questions**

1. The crystal structure adopted by iron is shown below. The distance between the nearest iron atoms is



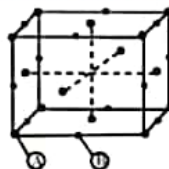
- a) 286 pm b) 124 pm c) 143 pm d) 247.6 pm
2. The body centred cubic cell of chromium has a length of 0.288 nm. Calculate the density of chromium (g/cm^3) : (Atomic weight : Cr = 52.0)
 a) 6.80 b) 7.60 c) 6.60 d) 7.26
3. First three nearest neighbour distance for body centered cubic lattice are
 a) $\sqrt{2}a, a, \sqrt{3}a$ b) $\frac{a}{\sqrt{2}}, a\sqrt{3}, a$ c) $\frac{\sqrt{3}a}{2}, a, \sqrt{2}a$ d) $\frac{\sqrt{3}a}{2}, a, \sqrt{3}a$

4. If 'a' stands for the edge length of the cubic systems: simple cubic, body centered cubic and face centred cubic, then the ratio of radii of the spheres in these systems will be respectively.

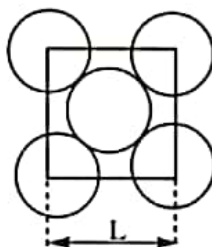
a) $\frac{a}{2} : \frac{a\sqrt{3}}{4} : \frac{a\sqrt{2}}{2}$ b) $a : \sqrt{3}a : \sqrt{2}a$ c) $\frac{a}{2} : \frac{a\sqrt{3}}{4} : \frac{a}{2\sqrt{2}}$ d) $\frac{a}{2} : \sqrt{3}a : \frac{a}{\sqrt{2}}$

5. For a solid with the following structure, the coordination number of the point B is

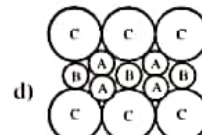
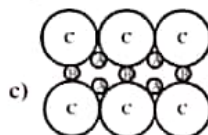
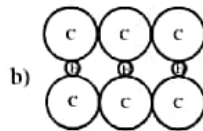
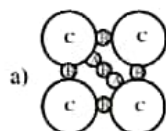
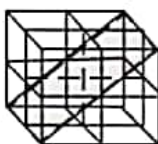
- a) 3
b) 4
c) 5
d) 6



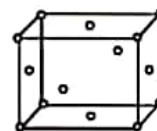
6. In rock salt structure, what percentage of the octahedral sites are occupied by cations?
a) 50% b) 33% c) 75% d) 100%
7. Determine the simplest formula of an ionic compound in which cations present at the corners and anions occur at the centre of each face
a) A_2B_3 b) AB_3 c) AB_2 d) AB_4
8. The packing efficiency of the two-dimensional square unit cell shown below is



- a) 39.27 % b) 68.02% c) 74.05% d) 78.54%
9. Calculate the number of ZnS units in a unit cell of zinc blende.
a) 6 b) 2 c) 4 d) 8
10. In a hypothetical solid C atoms are found to form cubical close packed lattice, A atoms occupy all tetrahedral voids while B atoms occupy all octahedral voids. A and B atoms are of appropriate size, so that there is no distortion in ccp lattice of C atoms. Now if a plane as shown in the following figure is cut, then the cross section of this plane will look like

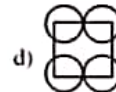
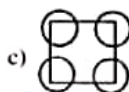


11. The intermediate compound LiAg crystallises in a cubic lattice in which both Li and Ag atoms have co-ordination number of 8. To what crystal class does the unit cell belong
 a) NaCl b) CsCl c) ZnS d) CsF₂
12. AgI crystallises in the cubic close-packed zinc blend structure. Assuming that the iodide ions occupy the lattice points, fraction of the tetrahedral sites occupied by silver ions are
 a) 50% b) 75% c) 100 % d) 33.3%
13. For the structure of solid given below if the lattice points represent A⁺ ions and the B⁻ ions occupy the tetrahedral voids then coordination number of A and B may be
 a) 2, 4 b) 4, 6
 c) 6, 4 d) 8, 4

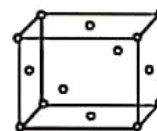


More than One correct answer Type Questions

14. The two types of holes which occur in any close-packed structures are
 a) tetrahedral, octahedral b) trigonal, octahedral
 c) trigonal, tetrahedral d) only octahedral
15. In face centered cubic structure, the octahedral voids are located at :
 a) edge centres b) body centre c) face centres d) corners
16. Select the correct statements among following: (a = edge length of unit cell)
 a) Nearest neighbour distance in NaCl = $\frac{a}{2}$
 b) Nearest neighbour distance in CaF₂ = $\frac{a\sqrt{3}}{4}$
 c) Nearest neighbour distance in Na₂O = $\frac{a\sqrt{3}}{4}$
 d) Nearest neighbour distance in CsCl = $\frac{a\sqrt{3}}{2}$
17. Which of the following statements are correct?
 a) The co-ordination number of each type of ions in CsCl crystals is 8.
 b) A metal which crystallizes in bcc structure has co-ordination number of 12.
 c) The length of a unit cell in NaCl is 552 pm. ($r_{Na^+} = 95\text{pm}$, $r_{Cl^-} = 181\text{pm}$)
 d) A unit cell of an ionic crystals shares some of its ions with other unit cells.
18. Which of the following statement(s) is (are) correct?
 a) When the radius ratio is in the range 0.414 - 0.732, a B.C.C arrangement with co-ordination no. 8
 b) When the radius ratio is in the range 0.225 - 0.414, a tetrahedral arrangement with co-ordination no. 4.
 c) When the radius ratio is in the range 0.155 - 0.225, an octahedral arrangement with co-ordination no. 6.
 d) In B₂O₃, smaller cations occupy triangular voids and a planar trigonal arrangement with co-ordination no. 3
19. Which planes can be found in a bcc unit cell?

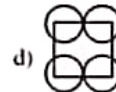
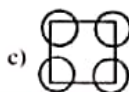


11. The intermediate compound LiAg crystallises in a cubic lattice in which both Li and Ag atoms have co-ordination number of 8. To what crystal class does the unit cell belong
 a) NaCl b) CsCl c) ZnS d) CsF_2
12. AgI crystallises in the cubic close-packed zinc blend structure. Assuming that the iodide ions occupy the lattice points, fraction of the tetrahedral sites occupied by silver ions are
 a) 50% b) 75% c) 100 % d) 33.3%
13. For the structure of solid given below if the lattice points represent A^+ ions and the B^- ions occupy the tetrahedral voids then coordination number of A and B may be
 a) 2, 4 b) 4, 6
 c) 6, 4 d) 8, 4



More than One correct answer Type Questions

14. The two types of holes which occur in any close-packed structures are
 a) tetrahedral, octahedral b) trigonal, octahedral
 c) trigonal, tetrahedral d) only octahedral
15. In face centered cubic structure, the octahedral voids are located at :
 a) edge centres b) body centre c) face centres d) corners
16. Select the correct statements among following: (a = edge length of unit cell)
 a) Nearest neighbour distance in NaCl = $\frac{a}{2}$
 b) Nearest neighbour distance in CaF_2 = $\frac{a\sqrt{3}}{4}$
 c) Nearest neighbour distance in Na_2O = $\frac{a\sqrt{3}}{4}$
 d) Nearest neighbour distance in CsCl = $\frac{a\sqrt{3}}{2}$
17. Which of the following statements are correct?
 a) The co-ordination number of each type of ions in CsCl crystals is 8.
 b) A metal which crystallizes in bcc structure has co-ordination number of 12.
 c) The length of a unit cell in NaCl is 552 pm. ($r_{\text{Na}^+} = 95\text{pm}$, $r_{\text{Cl}^-} = 181\text{pm}$)
 d) A unit cell of an ionic crystals shares some of its ions with other unit cells.
18. Which of the following statement(s) is (are) correct?
 a) When the radius ratio is in the range 0.414 - 0.732, a B.C.C arrangement with co-ordination no. 8
 b) When the radius ratio is in the range 0.225 - 0.414, a tetrahedral arrangement with co-ordination no. 4.
 c) When the radius ratio is in the range 0.155 - 0.225, an octahedral arrangement with co-ordination no. 6.
 d) In B_2O_3 , smaller cations occupy triangular voids and a planar trigonal arrangement with co-ordination no. 3
19. Which planes can be found in a bcc unit cell?



20. Which planes can be found in fluorite structure (unit cell)?



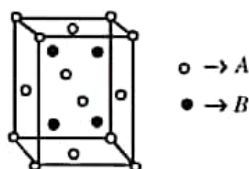
21. Which of the following statements is or are true?

- a) two ions A^+ and B^- have radii 88 pm and 200 pm respectively. In the close packed crystal of compound AB, the coordination number of A^+ is 6
 b) in CsCl crystal, edge length is 404 pm. The distance between the nearest neighbours is 350 pm.
 c) due to Frenkel defect, the density of the ionic solids does not changes
 d) the volume of atoms present in a face centered cubic unit cell of a metal (r is atom radii) is

$$\frac{16}{3}\pi r^3$$

Linked Comprehension Type Questions

Passage-I :



Answer the following questions for the above unit cell

22. If the molar mass of AB is 100 g mol^{-1} and 'a' is edge length then the density of the crystal will be

- a) $\frac{4N_A}{a^3 \times 100}$ b) $\frac{4 \times 100}{a^3 N_A}$ c) $\frac{2N_A}{a^3 100}$ d) $\frac{2 \times 100}{a^3 N_A}$

23. The given unit cell belongs to

- a) CsCl type b) TiCl type c) rock salt type d) zinc blende type

24. The coordination number of 'B' will be

- a) 8 b) 6 c) 4 d) 12

Passage-II :

O^{2-} : CCP

B^{3+} : Half of octahedral Voids

A^{2+} : $1/8^{\text{th}}$ of tetrahedral Voids

25. The space lattice described refers to

- a) fluorite structure b) rock salt structure c) spinel structure d) inverse spinel structure

26. The formula of the compound is

- a) ABO_2 b) A_2BO_3 c) AB_2O_4 d) A_2BO_4

27. Which of the following is an example of this structure

- a) $ZnAl_2O_4$ b) Zn_2AlO_4 c) $ZnAlO_2$ d) Zn_2AlO_3

Passage-III :

Following is the fact about the newly discovered superconductor of C_{60} (fullerene). The alkali metal fulleride superconductor M_3C_{60} has a cubic closest-packed (face-centered cubic) arrangement of nearly spherical C_{60}^{3-} anions with M^+ cations in the holes between the larger C_{60}^{3-} ions. The holes are of two types-octahedral holes, which are surrounded octahedrally by six C_{60}^{3-} ions, and tetrahedral holes, which are surrounded tetrahedrally by four C_{60}^{3-} ions.

28. How many C_{60}^{3-} ions, octahedral holes, and tetrahedral holes are present per unit cell
 a) 5,4,4 b) 3,8,4 c) 4,4,8 d) 2,1,2
29. Specify fractional coordinates for the tetrahedral holes (Fractional coordinates are fractions of the unit cell edge lengths. For example, a hole at the centre of the cell has fractional coordinates $\frac{1}{2}, \frac{1}{2}, \frac{1}{2}$)
 a) $\frac{1}{2}, \frac{1}{2}, \frac{1}{2}$ b) $\frac{1}{3}, \frac{1}{3}, \frac{1}{3}$ c) $\frac{1}{4}, \frac{1}{4}, \frac{1}{4}$ d) $\frac{1}{2}, \frac{1}{2}, \frac{1}{4}$
30. The ionic radii of Na^+ , K^+ and Rb^+ are 97, 133 and 147 pm, respectively. Which of these ions will fit into the octahedral holes? (radius of C_{60}^{3-} is about 350 pm)
 a) Na^+ b) K^+ c) Rb^+ d) all of these

Matrix Matching Type Questions

31. Match the following closest packing of identical spheres listed in Column-I with the characteristics listed in Column-II.

Column-I

- A) AAAA packing
 B) ABAB packing
 C) ABCABC packing
 D) Square close packing

Column-II

- p) CCP, CN = 12
 q) HCP, CN = 12
 r) BCC, CN = 8
 s) Primitive cubic, CN = 6

32. Match the elements (in Column-I) with the shape of the crystal (in Column-II)

Column-I

- A) Be
 B) Ca
 C) Ba
 D) Po

Column-II

- p) Body-centred cubic
 q) Simple cubic
 r) Face-centred cubic
 s) Hexagonal close-packed

33. **Column-I**

- A) 50% 'Tetrahedral' voids filled
 B) 100% 'octahedral' voids
 C) 100% 'Tetrahedral' voids filled
 D) No 'Tetrahedral' voids present

Column-II

- p) $CsCl$
 q) $NaCl$
 r) ZnS
 s) Na_2O

Integer Type Questions

34. Number of atoms per unit cell for body centered cubic system is ____
35. How many effective Cl^- ions are present in the rock salt $NaCl$ if ions along one axis joining opposite faces are removed?

36. A spinel is an important class of oxides consisting of two types of metal ions with oxide ions arranged in CCP layers. The normal spinel has $1/8^{\text{th}}$ of the tetrahedral void occupied by one type of metal and one half of the octahedral voids occupied by another type of metal ions such a spinel is formed by Zn^{2+} , Al^{3+} and O^{2-} with Zn^{2+} in tetrahedral void. Then the simplest formula of that spinel is $\text{Zn}_x\text{Al}_y\text{O}_z$ then $x + y + z$ is
37. A metal crystallizes in f.c.c. Then, the ratio of number of its first nearest neighbours to the second nearest neighbours is
38. Find the co-ordination number of sodium in Na_2O .

EXERCISE-III

(Defects and Properties of Solids)

LEVEL-I (MAIN)

Straight Objective Type Questions

- (A) : Thermodynamically all solids possess a tendency to acquire defects
(R) : During defects the entropy of the system increases in solids.
1) Both (A) and (R) are true and (R) is the correct explanation of (A)
2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
3) (A) is true but (R) is false
4) (A) is false but (R) is true
- At zero kelvin, most of the ionic crystals possess
1) Frenkel defect 2) Schottky defect 3) Metal excess defect 4) No defect
- In stoichiometric defects, the ratio of positive and negative ions as indicated by chemical formula of the compound
1) Decreases 2) Increases 3) Remains same 4) Cannot be predicted
- In a solid lattice the cation has left a lattice site and is located at an interstitial position, the lattice defect is
1) Metal excess defect 2) Vacancy defect 3) Frenkel defect 4) Schottky defect
- Non stoichiometric solid among the following
1) MgO 2) CaO 3) Na_2O 4) TiO
- Which of the following has both Schottky and Frenkel defects.
1) AgBr 2) ZnO 3) NaCl 4) KCl
- On doping Ge metal with a little of gallium one gets
1) p-type 2) n-type 3) Insulator 4) Rectifier
- Which substance will conduct the current in the solid state
1) Diamond 2) Graphite 3) Iodine 4) Sodium chloride
- An example for metallic conductor and semiconductor is
1) TiO 2) FeO 3) V_2O_3 4) NiO
- Molten sodium chloride conducts electricity due to the presence of
1) Free electrons 2) Free ions
3) Free molecules 4) Atoms of sodium and chlorine

11. Which of the following is ferromagnetic?
 1) Ni 2) Co 3) CrO_2 4) All
12. The oxide that is insulator is
 1) VO 2) MnO 3) ReO_3 4) Ti_2O_3
13. In which of the following the conductivity would be in the order of $10^{-4} \text{ ohm}^{-1} \text{ cm}^{-1}$
 1) $\text{NaCl}_{(s)}$ 2) $\text{Na}_{(s)}$ 3) diamond 4) Ge
14. (A) : Schottky and Frenkel defects are also called as 'thermo dynamic defects'.
 (R) : Both Schottky and Frenkel defects increase with increase in temperature.
15. Schottky defect causes
 1) Increase in the density of solid 2) Decrease in the density of solid
 3) No change in the density of solid 4) Decrease in the conductivity of solid
16. Which among the following is likely to have Schottky defect?
 1) AgCl 2) NaCl 3) TiCl 4) MgCl_2

Numerical Value Type Questions

17. Metals have conductivity in the order of 10^8 to 10^9 then $(x + y) = \underline{\hspace{1cm}}$ ($\text{ohm}^{-1} \text{ cm}^{-1}$)
18. To get n-type semiconductor, impurity to be added to silicon should have the how many number of valence electrons?

LEVEL-II (ADVANCED)**Straight Objective Type Questions**

1. Schottky-Wagner defects are mostly found in
 a) Ionic compounds with high co-ordination number
 b) Ionic compound with low co-ordination number
 c) Covalent compounds with low coordination number
 d) Covalent compound with high coordination number
2. Germanium can be made n-type semi conductor by doping with
 a) silicon b) arsenic c) gallium d) either As (or) Ga
3. (A) : Metals are generally good conductors of electricity.
 (R) : Electrical conductivity of metals is due to Schottky type defects.
 a) Both (A) and (R) are true and (R) is the correct explanation of (A)
 b) Both (A) and (R) are true and (R) is not the correct explanation of (A)
 c) (A) is true but (R) is false
 d) (A) is false but (R) is true
4. (A) : Antiferromagnetic substances possess almost zero magnetic moment.
 (R) : There are no unpaired electrons in anti-ferromagnetic substances.
 a) Both (A) and (R) are true and (R) is the correct explanation of (A)
 b) Both (A) and (R) are true and (R) is not the correct explanation of (A)
 c) (A) is true but (R) is false
 d) (A) is false but (R) is true

5. (A) : Fe_3O_4 is ferrimagnetic at room temperature but becomes paramagnetic at 850 K.
 (R) : The magnetic moments in Fe_3O_4 are aligned equally in parallel and antiparallel directions which on heating randomise.
 a) Both (A) and (R) are true and (R) is the correct explanation of (A)
 b) Both (A) and (R) are true and (R) is not the correct explanation of (A)
 c) (A) is true but (R) is false
 d) (A) is false but (R) is true
6. Frenkel defect appears in
 a) AgI
 b) ZnS
 c) AgBr
 d) all of these
7. An electron trapped in an anion vacancy within the crystal is called
 a) n-type conductor
 b) p-type conductor
 c) F-centre
 d) insulator
8. The p-type semiconductor is obtained when Si is doped with
 a) Sn
 b) Ge
 c) Ga
 d) As
9. The substance whose resistance gets reduced to virtually zero at very low temperature is called
 a) electrical conductor
 b) hyper conductor
 c) semiconductor
 d) super conductor
10. The alignment of magnetic dipoles in antiferro-magnetism is
 a) $\uparrow\uparrow\uparrow\uparrow$
 b) $\uparrow\downarrow\uparrow\downarrow$
 c) both (a) and (b)
 d) none of these
11. The formula of an oxide of iron is $\text{Fe}_{0.93}\text{O}_{1.00}$. If the compound has hundred O^{2-} ions, then it contains
 a) 93Fe^{+2} ions
 b) 93Fe^{+3} ions
 c) $79\text{Fe}^{+2}, 14\text{Fe}^{+3}$
 d) $93\text{Fe}^{+2}, 14\text{Fe}^{+3}$

More than One correct answer Type Questions

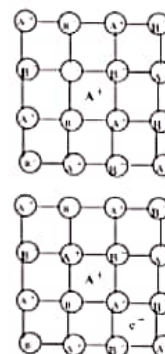
12. Which is true?
 a) Piezoelectricity is due to net dipole moment
 b) Some electric current is produced on heating polar crystals, this is pyroelectricity
 c) Ferroelectricity is due to alignment of dipoles in same direction
 d) Ferroelectricity is due to alignment of dipoles in opposite direction
13. The presence of F-centres in a crystal makes it
 a) conducting
 b) colourless
 c) coloured
 d) non-conducting
14. Which of the following is/are correct about the point defects?
 a) In Frenkel defect, the dielectric constant of solid increases
 b) In Schottky defect, the density of solid decreases
 c) In Frenkel defect, the density of solid decreases
 d) In Schottky defect, the dielectric constant of solid increases
15. Which of the following statements are correct about the diamagnetic solids?
 a) They have only paired electrons
 b) They are weakly repelled in magnetic field
 c) They have large number of unpaired electrons
 d) Chromium is diamagnetic
16. Which of the following statement(s) is/are correct?
 a) Frenkel defects occur when difference in size of cations and anions are bigger
 b) Schottky defects occur when cations and anions have similar ionic size
 c) An ionic crystal can have both Frenkel and Schottky defects
 d) Pure alkali metals do not have Frenkel defects

Linked Comprehension Type Questions

Passage-I :

Ionic lattice has two major point defects, (1) Schottky (2) Frenkel defects, Schottky defects occurs due to the cation - anion pairs missing from the lattice sites, Frenkel defects occur when an ion leaves its lattice site and fits into an interstitial space. The neutrality of the crystals is being maintained and we consider all losses from interstitial positions

17. Which defect decreases density of the crystal?
 a) Frenkel defect b) Schottky defect c) Both (a) and (b) d) None of these
18. Structure shown here represents
 a) Schottky defect
 b) Frenkel defect
 c) Metal excess defect
 d) None of these
19. Structure shown here represents :
 a) Schottky defect
 b) Frenkel defect
 c) Both defects
 d) None of these



Passage-II :

When a silicon crystal is doped with a group-15 element (with five valence electrons) such as P, As, Sb or Bi, the structure of the crystal lattice remains unchanged. Out of the five valence electrons of group-15 doped element, four electrons are used in normal covalent bonding with silicon and fifth electron is delocalised and thus conducts electricity. Doping a silicon crystal with a group-13 element (with three valence electrons) such as B, Al, Ga or In produces a semiconductor with three electrons in dopant. The place where fourth electron is missing is called as electron vacancy or hole.

20. Silicon that has been doped with group-15 elements is called
 a) p-type semiconductor b) n-type semiconductor
 c) electron vacancy or hole d) e-type semiconductor
21. Silicon that has been doped with group-13 elements is called
 a) p-type semiconductor b) n-type semiconductor
 c) electron vacancy or hole d) e-type semiconductor
22. If NaCl is doped with 10^{-3} mol% SrCl_2 , then concentration of cation vacancies is
 a) 6.02×10^{23} b) 6.02×10^{20} c) 6.02×10^{18} d) 6.02×10^{15}

Matrix Matching Type Questions

- | 23. Column-I (Electrical Properties) | Column-II (Materials) |
|--|--|
| A) Pure crystal of silicon at 0 K | p) Semi conductor – p-holes carry current |
| B) Pure crystal of silicon at 400 K | q) Semi conductor – electrons carry current |
| C) Silicon crystal doped with arsenic impurity | r) Insulator |
| D) Silicon crystal doped with gallium | s) Semiconductor-equal number of p-holes and electrons carry current |

24. Column-I (Solids)

- A) Piezoelectric
B) Antipiezoelectric
C) Ferroelectric
D) Pyroelectric

Column-II (Characteristics)

- p) Electric dipoles spontaneously aligned in one direction
q) Heating causes electric field
r) Mechanical stress causes electric field
s) Electric field causes elastic deformation

25. Column-I

- A) 'Na' vapour blown into NaCl
B) $\text{Fe}_{(1-x)}\text{O}$
C) Hot ZnO
D) Mixture of NaCl, SnCl_2

Column-II

- p) Doping
q) F-centres
r) Variable metal valency
s) Metal excess

Integer Type Questions

26. Analysis shows that nickel oxide has formula $\text{Ni}_{0.98}\text{O}_{1.00}$, the % of Ni^{3+} is x%. Then 'x' is
27. In $\text{Fe}_{0.96}\text{O}$, per one Avogadro number of oxide ions, $x \times 10^{-2}$ Avogadro number of cation vacancies are present. x value is

KEY SHEET (LECTURE SHEET)

EXERCISE-I

LEVEL-I

- 1) 4 2) 4 3) 4 4) 1 5) 2 6) 3 7) 3 8) 1
9) 1 10) 1 11) 1 12) 4 13) 2 14) 4 15) 3 16) 1.61
17) 1 18) 30

LEVEL-II

- 1) b 2) d 3) d 4) a 5) c 6) d 7) d 8) c
9) b 10) ab 11) acd 12) ac 13) cd 14) b 15) d
16) A-s; B-r; C-q; D-p 17) A-s; B-p; C-r; D-q 18) 3 19) 4

EXERCISE-II

LEVEL-I

- 1) 4 2) 4 3) 1 4) 3 5) 1 6) 4 7) 2 8) 3
9) 4 10) 3 11) 3 12) 3 13) 3 14) 3 15) 3 16) 3
17) 3 18) 1 19) 2 20) 2 21) 3 22) 0.15 23) 1.86 24) 0.41

LEVEL-II

- 1) d 2) d 3) c 4) c 5) d 6) d 7) b 8) d
9) c 10) c 11) b 12) a 13) d 14) abc 15) ab 16) abcd
17) acd 18) bd 19) abc 20) abd 21) abcd 22) b 23) d 24) c
25) c 26) c 27) a 28) c 29) c 30) c
31) A-s; B-q; C-p; D-s 32) A-s; B-r; C-p; D-q
33) A-r; B-q; C-s; D-p 34) 2 35) 3 36) 7 37) 2 38) 4

EXERCISE-III

LEVEL-I

- 1) 1 2) 4 3) 3 4) 3 5) 4 6) 1 7) 1 8) 2
9) 3 10) 2 11) 4 12) 2 13) 4 14) 1 15) 2 16) 2
17) 7 18) 5

LEVEL-II

- 1) a 2) b 3) c 4) c 5) c 6) d 7) c 8) c
9) d 10) b 11) c 12) abc 13) ac 14) ab 15) ab 16) abcd
17) b 18) b 19) d 20) b 21) a 22) c
23) A-r; B-s; C-q; D-p 24) A-r; B-s; C-p; D-q
25) A-qs; B-r; C-qs; D-p 26) 4 27) 4